Experiment-9:

Compare Linear and Polynomial Regression using Python

Code:

```
# Compare Linear Regression and Polynomial Regression
# Student Performance Dataset
from google.colab import files
uploaded = files.upload()
import pandas as pd
import io
# Read CSV
filename = list(uploaded.keys())[0]
df = pd.read_csv(io.BytesIO(uploaded[filename]))
print("First 5 Rows:")
print(df.head())
# Encode categorical columns
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])
# Features and Target
X = df.drop("math score", axis=1)
y = df["math score"]

# Train-Test Split
```

```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# ---------------------------
# Linear Regression
# ---------------------------
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)
# ---------------------------
# Polynomial Regression
# ---------------------------
from sklearn.preprocessing import PolynomialFeatures
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)
poly_model = LinearRegression()

poly_model.fit(X_train_poly, y_train)
y_pred_poly = poly_model.predict(X_test_poly)
```

```
# ----------------------------
# Evaluation
# ----------------------------
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
print("\n========== Linear Regression ==========")
print("MAE :", mean_absolute_error(y_test, y_pred_lr))
print("MSE :", mean_squared_error(y_test, y_pred_lr))
print("R2 Score :", r2_score(y_test, y_pred_lr))
print("\n========== Polynomial Regression ==========")
print("MAE :", mean_absolute_error(y_test, y_pred_poly))
print("MSE :", mean_squared_error(y_test, y_pred_poly))
print("R2 Score :", r2_score(y_test, y_pred_poly))
# Comparison Table
comparison = pd.DataFrame({
    "Model": ["Linear Regression", "Polynomial Regression"],
    "MAE": [
        mean_absolute_error(y_test, y_pred_lr),
        mean_absolute_error(y_test, y_pred_poly)
    ],
    "MSE": [
        mean_squared_error(y_test, y_pred_lr),
        mean_squared_error(y_test, y_pred_poly)
    ],
```

```
    "R2 Score": [
        r2_score(y_test, y_pred_lr),
        r2_score(y_test, y_pred_poly)
    ]
})
print("\nComparison:")
```

print(comparison)

Output:

Choose Files StudentsPe...ance[1].csv

StudentsPerformance[1].csv(text/csv) - 72036 bytes, last modified: 8/4/2026 - 100% done

```
Saving StudentsPerformance[1].csv to StudentsPerformance[1].csv
First 5 Rows:
   gender race/ethnicity parental level of education         lunch  \
0  female        group B           bachelor's degree      standard
1  female        group C                some college      standard
2  female        group B             master's degree      standard
3    male        group A          associate's degree  free/reduced
4    male        group C                some college      standard

  test preparation course  math score  reading score  writing score
0                    none          72             72             74
1               completed          69             90             88
2                    none          90             95             93
3                    none          47             57             44
4                    none          76             78             75

========== Linear Regression ==========
MAE : 4.1301450780040545
MSE : 28.275284506327335
R2 Score : 0.8838026201112223

========== Polynomial Regression ==========
MAE : 4.403727404217777
MSE : 31.813800541995942
R2 Score : 0.8692610761721292

Comparison:
                   Model       MAE        MSE  R2 Score
0      Linear Regression  4.130145  28.275285  0.883803
1  Polynomial Regression  4.403727  31.813801  0.869261
```